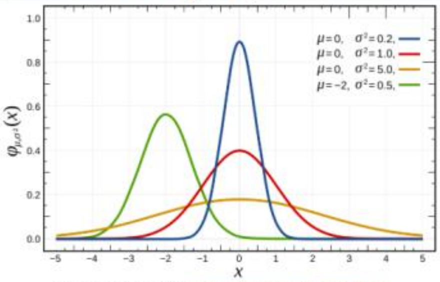
Standard Normal Distribution

08 June 2026 14:57

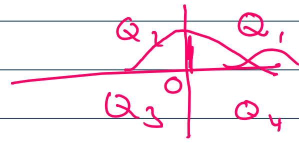
Normal distribution

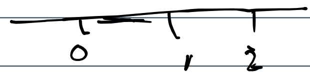
Probability density function



The red curve is the *standard normal distribution*.

$\mu \Rightarrow 0$.





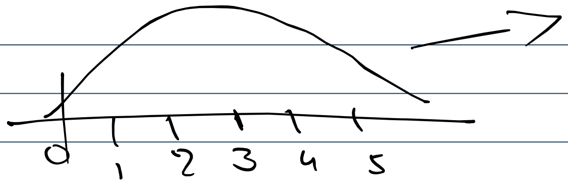
Standard normal distribution [edit]

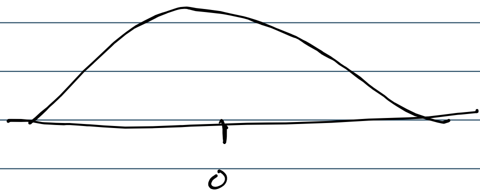
The simplest case of a normal distribution is known as the **standard normal distribution** or **unit normal distribution**. This is a special case when $\mu = 0$ and $\sigma^2 = 1$, and it is described by this probability density function (or density):[11]

$x = \{1, 2, 3, 4, 5\}$ $\quad \mu = 3$

$\Rightarrow$

$\sigma = 1.414 \approx 1$





$$z\text{-score} = \frac{x_i - \mu}{\sigma}$$

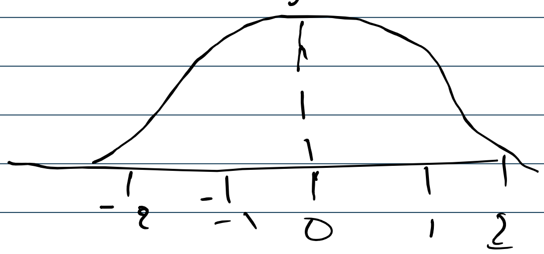
$x = \{1, 2, 3, 4, 5\} \Rightarrow y = \{-2, -1, 0, 1, 2\}$

1) $\frac{1-3}{1} \Rightarrow -2$ $\quad$ 4) $\Rightarrow \frac{4-3}{1} = 1$

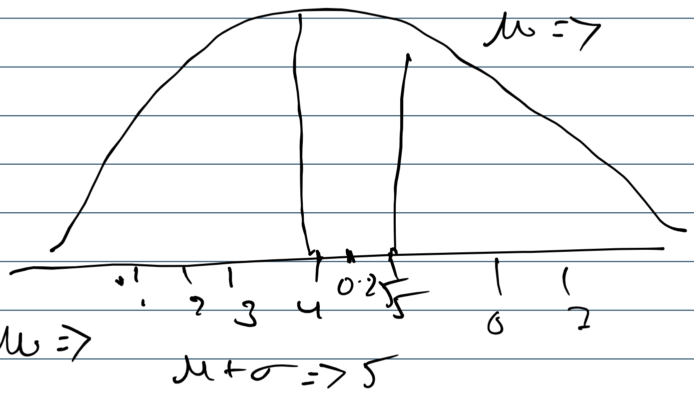
2) $\frac{2-3}{1} \Rightarrow -1$ $\quad$ 5) $\Rightarrow \frac{5-3}{1} \Rightarrow 2$

3) $\frac{3-3}{1} \Rightarrow 0$

$\mu \quad \sigma = 1$



$$x \approx SND(\mu = 0, \sigma = 1)$$



How many st-d 4.25 is away from the mean

$$z\text{-score} = \frac{x-\mu}{\sigma}$$

σ^2

$\Downarrow$

σ

$$= \frac{4.25 - 4.0}{1}$$

$$= 0.25$$

$x_i : 2.5 \Rightarrow \mu$

$$= \frac{2.25 - 4}{1} \Rightarrow -1.5$$

Eg: Dataset

Age	Weight	Heigh	Salary
20	60	165	40K
21	70	180	30K
22	80	190	40K
34	90	205 cm	70K
$\Downarrow$ year	$\Downarrow$ kgs	$\Downarrow$ cm $\Rightarrow$ m $\Rightarrow$ ft	$\Downarrow$ Thr

units $\Rightarrow$ All of unit $\Rightarrow$

① Standardization.